Building Energy Benchmark: Computer Laboratories High-Density Compute Audit

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BUILDING ENERGY BENCHMARK: COMPUTER LABORATORIES HIGH-DENSITY COMPUTE AUDIT

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Facility Code: BLD_LAB (Computer Laboratories Building)

Departmental Jurisdiction: Department of Computer Engineering (DEP_CMPN) and Department of Artificial Intelligence & Data Science (DEP_AIDS).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. FACILITY ARCHITECTURE AND COMPUTE ASSET INVENTORY

The Computer Laboratories building (BLD_LAB) encompasses 5,200 m2 of specialized academic floor area across four levels. The facility accommodates over 650 concurrent students and faculty, housing 16 dedicated departmental computer laboratories, two enterprise server rooms, and an Advanced AI & High-Performance Computing Research Center. Compute infrastructure includes 680 desktop workstations, 12 GPU-accelerated machine learning workstation towers (NVIDIA RTX A6000 clusters), and two enterprise server rack cabinets running round-the-clock academic services, repository mirrors, and campus AI workloads.

2. ENERGY CONSUMPTION ANALYSIS AND BENCHMARK DISPARITY

In calendar year 2024, facility sub-metering under project INIT_IOT_ENERGY_MGT registered a gross annual electricity consumption of 582,400 kWh for BLD_LAB. Despite representing only 16.0% of the institution's total built-up area (32,500 m2), the building accounts for 24.65% of total campus electrical consumption (2,362,400 kWh).

- Specific Energy Consumption (SEC): Measured at 112.0 kWh/m2/year. This is more than double the specific energy intensity of Academic Block A (BLD_ACA at 51.0 kWh/m2/year) and Academic Block B (BLD_ACB at 49.0 kWh/m2/year).

- Carbon Attribution: At regional grid emission factor EF_ELEC_GRID (0.82 kg CO2e/kWh), the facility generated 477,568 kg CO2e in 2024, making it the single largest point source of Scope 2 emissions on campus.

3. SUB-SYSTEM LOAD DISTRIBUTION AND LOSS ANALYSIS

Detailed power distribution monitoring identified four core consumption streams:

- IT Workstations and Compute Hardware: 44.0% (256,256 kWh/year). Lab PCs and GPU workstations draw substantial energy during daily sessions. However, 18.2% of workstation power was consumed during idle evening and weekend hours due to inactive PCs left plugged into unswitched bench outlets.

- Precision HVAC and Space Air Conditioning: 41.0% (238,784 kWh/year). Space cooling is provided by a combination of conventional split air conditioners in teaching labs and dedicated precision air handling units (PAHUs) in server rooms.

- Architectural LED Lighting: 9.0% (52,416 kWh/year).

- UPS Conversion Inefficiencies and Auxiliary Losses: 6.0% (34,944 kWh/year). Central double-conversion online UPS units operate at an average load factor of only 38%, resulting in an 11% parasitic thermal conversion loss.

4. SERVER ROOM POWER USAGE EFFECTIVENESS (PUE) AUDIT

The audit evaluated the primary AI Research Server Room housing campus computing clusters:

- Total Server Room Power Draw: 48,200 kWh/year.

- IT Equipment Compute Energy: 27,700 kWh/year.

- Server Room Power Usage Effectiveness (PUE): Calculated at 1.74 (Total Power / IT Power). A PUE of 1.74 indicates significant cooling inefficiency, well above modern green data center benchmarks (1.20 - 1.35). Inefficiencies stem from mixing hot exhaust air with cold supply air across server racks.

5. TARGETED RETROFIT ACTION PLAN

To decarbonize BLD_LAB and support institutional targets TRG_NET_ZERO and TRG_RENEWABLE:

- Cold Aisle Containment: Install physical acrylic aisle doors and ceiling baffles in the server room to prevent thermal air mixing, coupled with adjusting cooling setpoints to 24.0°C. This will reduce PUE from 1.74 to 1.32, saving 16,500 kWh annually.

- Smart Automated Workstation Relays: Program automated master circuit contactors to de-energize lab benches between 20:30 and 07:30, eliminating 32,000 kWh/year of standby draw.

- Combined Savings: 48,500 kWh/year (39,770 kg CO2e avoided), supporting United Nations SDG 7, SDG 9, and SDG 12.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_BLD_011**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*